

POWER BI DAX CHEAT SHEET

Core Functions Grouped by Category, with Syntax

1. Mathematical & Aggregation Functions

```
-- SUM
Total Sales = SUM(Sales[SalesAmount])

-- SUMX
Total Sales = SUMX(Sales, Sales[Quantity] * Sales[UnitPrice])

-- AVERAGE
Average Sales = AVERAGE(Sales[SalesAmount])

-- AVERAGEX
Average Sales = AVERAGEX(Sales, Sales[Quantity] * Sales[UnitPrice])

-- MIN
Minimum Sales = MIN(Sales[SalesAmount])

-- MAX
Maximum Sales = MAX(Sales[SalesAmount])

-- COUNT
Employee Count = COUNT(Employee[EmployeeID])

-- COUNTA
Employee Count = COUNTA(Employee[EmployeeName])

-- COUNTROWS
Total Rows = COUNTROWS(Sales)

-- DISTINCTCOUNT
Customer Count = DISTINCTCOUNT(Sales[CustomerID])

-- DIVIDE
Profit Margin = DIVIDE([Profit], [Total Sales], 0)

-- ROUND
Rounded Value = ROUND([Total Sales], 2)

-- ABS
Absolute Value = ABS([Profit])

-- MOD
Remainder = MOD(Sales[Quantity], 2)
```

2. Logical Functions

```
-- IF
Status =
IF(
    Sales[SalesAmount] >= 10000,
```

```

    "High",
    "Low"
)

-- SWITCH
Grade =
SWITCH(
    TRUE(),
    [Marks] >= 90, "A+",
    [Marks] >= 80, "A",
    [Marks] >= 70, "B",
    [Marks] >= 60, "C",
    "Fail"
)

-- AND
Result =
IF(
    Sales[SalesAmount] > 10000 &&
    Sales[Quantity] > 10,
    "Yes",
    "No"
)

-- OR
Result =
IF(
    Sales[SalesAmount] > 10000 ||
    Sales[Quantity] > 10,
    "Yes",
    "No"
)

-- NOT
Result =
IF(
    NOT(Sales[Status] = "Cancelled"),
    "Valid",
    "Invalid"
)

-- IFERROR
Result =
IFERROR(
    [Sales] / [Quantity],
    0
)

-- COALESCE
Result =
COALESCE(
    [Sales],
    0
)

```

3. Text Functions

```
-- CONCATENATE
Full Name =
CONCATENATE(
    Employee[FirstName],
    Employee[LastName]
)

-- Concatenation operator
Full Name =
Employee[FirstName] & " " & Employee[LastName]

-- LEFT
First 3 Characters =
LEFT(Employee[EmployeeName], 3)

-- RIGHT
Last 3 Characters =
RIGHT(Employee[EmployeeName], 3)

-- MID
Middle Text =
MID(Employee[EmployeeName], 2, 4)

-- LEN
Name Length =
LEN(Employee[EmployeeName])

-- UPPER
Upper Name =
UPPER(Employee[EmployeeName])

-- LOWER
Lower Name =
LOWER(Employee[EmployeeName])

-- TRIM
Clean Name =
TRIM(Employee[EmployeeName])

-- SUBSTITUTE
Updated Name =
SUBSTITUTE(
    Employee[EmployeeName],
    "Old",
    "New"
)

-- SEARCH
Position =
SEARCH(
    "A",
    Employee[EmployeeName],
    1,
```

```

    0
)

-- FORMAT
Formatted Sales =
FORMAT(
    [Total Sales],
    "#,##0.00"
)

```

4. Date & Time Functions

```

-- TODAY
Today Date = TODAY()

-- NOW
Current DateTime = NOW()

-- YEAR
Sales Year = YEAR(Sales[OrderDate])

-- MONTH
Sales Month = MONTH(Sales[OrderDate])

-- DAY
Sales Day = DAY(Sales[OrderDate])

-- QUARTER
Sales Quarter = QUARTER(Sales[OrderDate])

-- WEEKDAY
Day Number = WEEKDAY(Sales[OrderDate])

-- WEEKNUM
Week Number = WEEKNUM(Sales[OrderDate])

-- DATE
New Date =
DATE(
    2026,
    9,
    11
)

-- DATEDIFF
Days Taken =
DATEDIFF(
    Sales[OrderDate],
    Sales[DeliveryDate],
    DAY
)

-- EOMONTH
Month End =

```

```

EOMONTH(
    Sales[OrderDate],
    0
)

-- EDATE
Next Month =
EDATE(
    Sales[OrderDate],
    1
)

```

5. Filter Functions

CALCULATE — ★ Very Important

```

Total Sales =
CALCULATE(
    SUM(Sales[SalesAmount])
)

```

With a filter:

```

Electronics Sales =
CALCULATE(
    [Total Sales],
    Product[Category] = "Electronics"
)

```

Multiple filters:

```

High Value Sales =
CALCULATE(
    [Total Sales],
    Product[Category] = "Electronics",
    Sales[SalesAmount] > 10000
)

```

FILTER

```

High Value Sales =
CALCULATE(
    [Total Sales],
    FILTER(
        Sales,
        Sales[SalesAmount] > 10000
    )
)

```

ALL

```

Total Sales All Products =
CALCULATE(
    [Total Sales],

```

```
    ALL(Product)
)
```

REMOVEFILTERS

```
Total Sales =
CALCULATE(
    [Total Sales],
    REMOVEFILTERS(Product)
)
```

ALLSELECTED

```
Selected Sales =
CALCULATE(
    [Total Sales],
    ALLSELECTED(Product)
)
```

VALUES

```
Selected Category =
VALUES(Product[Category])
```

SELECTEDVALUE — ★ Important

```
Selected Category =
SELECTEDVALUE(
    Product[Category],
    "All Categories"
)
```

6. Iterator Functions — X Functions

```
-- SUMX
Total Amount =
SUMX(
    Sales,
    Sales[Quantity] * Sales[UnitPrice]
)

-- AVERAGEX
Average Amount =
AVERAGEX(
    Sales,
    Sales[Quantity] * Sales[UnitPrice]
)

-- MINX
Minimum Amount =
MINX(
    Sales,
    Sales[Quantity] * Sales[UnitPrice]
)
```

```

-- MAXX
Maximum Amount =
MAXX(
    Sales,
    Sales[Quantity] * Sales[UnitPrice]
)

-- COUNTX
Product Count =
COUNTX(
    Sales,
    Sales[ProductID]
)

-- CONCATENATEX
Product List =
CONCATENATEX(
    VALUES(Product[ProductName]),
    Product[ProductName],
    ", "
)

```

7. Relationship Functions

RELATED

Used to retrieve a value from a related table.

```

Category =
RELATED(
    Product[Category]
)

```

RELATEDTABLE

```

Sales Count =
COUNTROWS(
    RELATEDTABLE(Sales)
)

```

USERELATIONSHIP

Used to activate an inactive relationship.

```

Sales by Delivery Date =
CALCULATE(
    [Total Sales],
    USERELATIONSHIP(
        Sales[DeliveryDate],
        DateTable[Date]
    )
)

```

8. Ranking Functions

RANKX — ★ Important

```
Product Rank =  
RANKX(  
    ALL(Product[ProductName]),  
    [Total Sales],  
    ,  
    DESC  
)
```

TOPN

```
Top 5 Products =  
TOPN(  
    5,  
    Product,  
    [Total Sales],  
    DESC  
)
```

9. Time Intelligence Functions

DATEADD

```
Previous Month Sales =  
CALCULATE(  
    [Total Sales],  
    DATEADD(  
        DateTable[Date],  
        -1,  
        MONTH  
    )  
)
```

SAMEPERIODLASTYEAR

```
Previous Year Sales =  
CALCULATE(  
    [Total Sales],  
    SAMEPERIODLASTYEAR(  
        DateTable[Date]  
    )  
)
```

TOTALYTD

```
YTD Sales =  
TOTALYTD(  
    [Total Sales],  
    DateTable[Date]  
)
```

TOTALMTD

```
MTD Sales =  
TOTALMTD(  
    [Total Sales],  
    DataTable[Date]  
)
```

TOTALQTD

```
QTD Sales =  
TOTALQTD(  
    [Total Sales],  
    DataTable[Date]  
)
```

PREVIOUSMONTH

```
Previous Month Sales =  
CALCULATE(  
    [Total Sales],  
    PREVIOUSMONTH(  
        DataTable[Date]  
    )  
)
```

PREVIOUSYEAR

```
Previous Year Sales =  
CALCULATE(  
    [Total Sales],  
    PREVIOUSYEAR(  
        DataTable[Date]  
    )  
)
```

10. Information Functions

```
-- ISBLANK  
Check Blank =  
IF(  
    ISBLANK([Total Sales]),  
    "Blank",  
    "Available"  
)  
  
-- ISNUMBER  
Check Number =  
ISNUMBER(Sales[SalesAmount])  
  
-- ISTEXT  
Check Text =  
ISTEXT(Employee[EmployeeName])
```

```
-- ISERROR
Check Error =
ISERROR(
    [Total Sales] / [Quantity]
)

-- HASONEVALUE
Check One Value =
HASONEVALUE(
    Product[Category]
)

-- ISFILTERED
Check Filter =
ISFILTERED(
    Product[Category]
)
```

